PULLED OUT OF THE FORTY

The thirteen that need a program

The other twenty-seven run on a program you already have, or on Port View with no program at all. These thirteen do not — so they are here on their own, with the same lesson numbers as the main document.

They are thirteen lessons, but only six programs

P1 alone covers seven of them. Build the six once and everything else is changing one number, or running two of them one after the other. Three lessons chain two programs; nothing needs a seventh.

- **P1** 7 Drive until it sees something
- **P2** 2 A list of timed moves
- **P3** 2 Hold a distance
- **P4** 1 Speedometer
- **P5** 2 Drive by remote
- **P6** 2 Turn until it finds the beacon

Written as logic, not as blocks

The steps are in plain language rather than EV3 block names, because the palette differs between EV3 Lab and EV3 Classroom. **Any of these is three to six blocks** in either one.

Where the programming actually sits

Eight of the thirteen are in **Control and feedback** — which makes sense, because that is the principle about a robot checking instead of assuming, and checking is the thing you have to write down. The sensing and measurement lessons are nearly all Port View.

BUILD THESE ONCE

The six programs

In the order worth building them. P1 is the one everything else leans on.

P1 Drive until it sees something

LESSONS 6, 8, 12, 13, 34, 38, 39

Three blocks. **Build this one first** — seven of the thirteen lessons are this program with one number changed.

```
Start both drive motors going forward.
Wait until the distance sensor reads less than 10.
Stop both motors.
```

8 runs it at 25%, 50% and 100% power. **12** runs it at each gear ratio. **13** tunes the trigger number until it lands. **34** subtracts the gap between the sensor face and the bumper.

P2 A list of timed moves

LESSONS 6, 10

No sensor at all. Just moves in order, each for a set time.

```
Forward for 2 seconds.
Turn for half a second.
Forward for 1 second.
Stop.
```

10 is whatever the driver actually did, written down and typed in — so the list is different every time.

P3 Hold a distance

LESSONS 7, 9

A loop that keeps checking and corrects. This is the first real feedback program they write.

```
Repeat for ever:
  If the distance is more than 23 → drive forward.
  If it is less than 17 → drive backward.
  Otherwise → stop.
```

The 23 and 17 are a target of 20, give or take 3. **9** points the sensor sideways and steers gently instead of driving straight.

P4 Speedometer

LESSONS 27

Two readings and a clock. Nothing on the robot measures speed, so this works it out.

```
Repeat for ever:
  Read the distance. Call it FIRST.
  Wait half a second.
  Read the distance again. Call it SECOND.
  Show (FIRST − SECOND) ÷ 0.5 on the screen.
```

Only works driving straight at a wall. Turn away and the number goes nonsense — which is worth letting them discover.

P5 Drive by remote

LESSONS 11, 39

One line per button. Long, but every line is the same shape.

```
Repeat for ever:
  If top-left is pressed → both motors forward.
  If bottom-left → both motors back.
  If top-right → turn right. If bottom-right → turn left.
  If nothing is pressed → stop.
```

39 adds one more line: if the center button is pressed, stop listening to the remote and run **P1** instead.

P6 Turn until it finds the beacon

LESSONS 37, 38

Spin slowly until the robot is pointing at the beacon, then stop.

```
Repeat until the beacon heading is between −2 and +2:
  Turn slowly on the spot.
Stop.
```

38 runs **P6** and then **P1**, one after the other. That is the whole lesson — nothing new gets written.

The three that chain two

Lesson 6 runs P2 and then P1, so the class can see the same job done blind and then done with the sensor. **Lesson 38** is P6 then P1. **Lesson 39** is P5 then P1, with a button to switch. **None of the three needs anything new written.**

6 Blind versus seeing

P1 + P2 · G·U · THE ROBOT · HANDS ●●○ · HEAD ●○○

INTENT A timed drive has the room memorized. A sensor reads the room as it actually is. This is the entire reason the sensor is on the robot.

SET-UP A wall or a large book stood on end, and a floor you can mark with tape.

THE TEST Write program A: drive forward for 3 seconds, stop. Tune the seconds until it stops a hand's width from the wall. Now move the wall 30 cm further back and run the same program again. Then write program B: drive forward *until* the ultrasonic reads under 10 cm, then stop. Move the wall anywhere you like and run it again.

OUTCOME Program A stops in the middle of nowhere — or, if you moved the wall closer, hits it. Program B stops 10 cm from the wall every single time, wherever you put it. **A timed or rotation-counted drive contains a hidden assumption about how the world is arranged, and it is wrong the moment anything moves.** Worth 10 minutes on day one.

7 Keep your distance

P3 · G·U · THE ROBOT · HANDS ●●○ · HEAD ●●○

INTENT Feedback control in its simplest possible form, and why a correction band that is too tight makes a robot shake instead of settle.

SET-UP A clipboard or book you hold in front of the robot and move by hand.

THE TEST Program a loop: if the distance is more than 20 cm, drive forward; if less than 20 cm, back up; otherwise stop. Run it, then walk the board slowly forwards and backwards. Then narrow the band from ±3 cm to ±1 cm and run it again.

OUTCOME The robot chases the board like a nervous dog. Wide band, it settles; narrow band, it judders endlessly, because each correction carries it straight out the other side. **That juddering is the oldest problem in control engineering and they just built it in four blocks.** The fix is not a better band but a correction that gets gentler as you approach — which is what a PID controller is.

8 Reaction distance

P1 · G·U · THE ROBOT · HANDS ●●○ · HEAD ●●○

INTENT Sensing takes time and stopping takes time. Everything that happens in that gap is distance you did not ask for.

SET-UP A wall, a tape measure, and the stop-at-20 cm program from lesson 6 or 7.

THE TEST Set the program to stop when the sensor reads 20 cm. Drive at the wall from 1 meter away. Measure where the robot *actually* ends up with a real tape. Repeat at 25%, 50% and 100% motor power, three runs each.

OUTCOME It never stops at 20. At low power it may stop at 18 cm; at full power, 12 cm or closer. **Three delays stack up: the sensor samples only every few hundredths of a second, the program loop takes time to come round again, and the robot has momentum.** This is braking distance, and it is why speed limits exist.

9 Wall follower

P3 · G·U · THE ROBOT · HANDS ●●● · HEAD ●●○

INTENT Overcorrection made visible: every correction is applied to a robot that has already moved somewhere else.

SET-UP A long straight wall, or a row of books lined up along the floor.

THE TEST Mount the ultrasonic pointing sideways. Drive forward continuously; if the gap to the wall is too big, steer gently towards it, if too small, steer away. Send it down a 3-meter wall.

OUTCOME It weaves down the wall in a slow S instead of tracking straight, and steering harder makes the weave worse rather than better. **Same deadband as lesson 7, except now the robot is moving, so every correction arrives late.** This is why ships zigzag and why learner drivers weave.

10 You were the program

P2 · G·IR · THE REMOTE · HANDS ●●○ · HEAD ●●●

INTENT The gap between what a driver does and what they can write down is exactly the thing a sensor puts back. If you run one lesson to explain why sensors exist, run this one.

SET-UP A simple course, a clipboard, and two students.

THE TEST One drives by remote while the other writes down every single action in order: forward two seconds, left a bit, forward, stop. Then type that exact list into the EV3 as a program and run it with nobody touching the remote.

OUTCOME It gets partway and goes wrong — drifting wide on a turn, stopping in the wrong place. The list looked complete and was not. **What is missing is everything the driver did with their eyes.** They were correcting continuously against what they saw, and never wrote it down because they never noticed doing it.

11 Piloted versus programmed

P5 · G·U·IR · BOTH SENSORS · HANDS ●●○ · HEAD ●●○

INTENT Autonomy does not buy speed. It buys repeatability — which is a different virtue, and the reason competitions score an autonomous period separately.

SET-UP A short course: forward, round a box, stop in a taped square.

THE TEST Run it twice over. First a student drives by remote, timed. Then a written program does the same course, timed. Five runs of each, recording every time rather than just the best.

OUTCOME The human usually beats the program on the single fastest run. But the human's five times are spread over several seconds and the program's five are nearly identical. **Have them write down both the average and the spread — the spread is the real finding**, and it is the number most teams never think to collect.

12 The overshoot curve

P1 · G·U · THE ROBOT · HANDS ●●● · HEAD ●●●

INTENT There is no best gear ratio. There is only the best ratio for a stated precision — which means the question "which gears?" cannot be answered on its own.

SET-UP Lesson 4's setup, plus two more gear ratios on the base.

THE TEST Run lesson 8 at three different gear ratios — fast, middle, deep reduction. Five runs each. For each, record the average error (how far past 20 cm it ended) and the spread (best run to worst run). Plot error against ratio on paper.

OUTCOME Fast gearing has a large error *and* a large spread — not just wrong, but wrong by a different amount each time. Deep reduction is accurate, consistent and slow. **This is the Week 3 trade-off with a number attached.** When someone asks in December which gears to use, the honest reply is: how accurately do you need to stop?

13 Slow and precise

P1 · G·U · THE ROBOT · HANDS ●●○ · HEAD ●○○

INTENT Precision is purchasable and the currency is time. They have already felt the fast end of this trade; here they feel the other end.

SET-UP A wall, a real tape measure, and the deepest reduction they can build.

THE TEST Target: stop at 20 cm from the wall, judged by the tape and not by the robot. Five consecutive runs. A run is a hit if it lands between 19 and 21 cm. Then try to beat it with a faster gearing.

OUTCOME Deep reduction usually hits 5 from 5. Fast gearing scatters across 8 cm and cannot be tuned out of it however they adjust the trigger. **Notice the trigger distance they settled on is not 20** — they compensated for lesson 8's reaction distance without being told to, which is worth saying out loud to them.

27 Build a speedometer

P4 · G·U · THE ROBOT · HANDS ●●● · HEAD ●●○

INTENT Speed is not something the robot can detect. It is something it calculates, from two positions and a clock.

SET-UP A clear run of two meters at a wall. Nothing else.

THE TEST Loop: read the distance, wait half a second, read it again, subtract, divide by 0.5. Show the answer on the brick screen in cm per second. Drive slowly at a wall and watch the number. Then drive at full power.

OUTCOME A live speed readout that rises and falls as they drive. It only works heading straight at the wall — turn away and the number goes nonsense. **Nothing on this robot measures speed.** Every speedometer ever built does this same subtraction, including the one in your car.

34 Stop on the line

P1 · G·U · THE ROBOT · HANDS ●●○ · HEAD ●●○

INTENT The sensor measures from its own face, not from the part of the robot you care about. Every reading needs an offset before it means anything useful.

SET-UP A tape line on the floor and a block or book standing on the line as a target to see.

THE TEST Challenge: stop with the robot's *front edge* exactly on the line. They will write "drive until distance < 0" and discover it cannot. Have them measure the gap between the sensor face and the front bumper — call it the offset — and stop at that number instead.

OUTCOME First attempts stop short or run the block over. Once they subtract the offset, it lands on the line; fast gearing still overshoots by a centimeter or two. **Every real robot has this problem — the camera is never where the gripper is.** Converting between where the sensor is and where the robot is is a permanent tax, not a beginner's mistake.

37 Beacon hunt

P6 · U·IR · BOTH SENSORS · HANDS ●●● · HEAD ●●○

INTENT Different sensors answer different questions — *which way* versus *how far* — and a robot needing both answers needs both sensors.

SET-UP The IR beacon hidden somewhere in the room at robot height, switched to channel 1.

THE TEST Program the robot to spin slowly until the IR seeker's heading value is near zero, then stop and drive forward. Let it hunt. Then ask them to make it stop 15 cm short of the beacon — which they will find they cannot do with infrared alone.

OUTCOME It turns to face the beacon reliably and sets off. Stopping at a set distance is hopeless until they add the ultrasonic to do the ranging. **Neither sensor can do the other's job, and the robot needs both jobs done.** The grown-up name for combining them is sensor fusion.

38 Search and approach

P1 + P6 · G·U·IR · BOTH SENSORS · HANDS ●●● · HEAD ●●○

INTENT A complete robot behavior — find, approach, stop — assembled entirely from parts they already understand. Nothing new is required.

SET-UP The beacon placed anywhere in the room, moved between runs.

THE TEST Chain three behaviors they have already built: spin until the IR heading is near zero, drive forward, stop when the ultrasonic reads under 15 cm. Place the beacon somewhere new and let it go.

OUTCOME The robot sweeps, locks on, crosses the room and parks in front of the beacon — from anywhere in the room, which is what makes it feel different from everything before it. **Three familiar pieces, combined, produce something that looks like a real robot.** A good last session before a break, and a good one to show parents.

39 The handoff

P1 + P5 · G·U·IR · BOTH SENSORS · HANDS ●●● · HEAD ●●○

INTENT Humans and programs are good at different parts of the same task. Splitting the task along that line is how nearly every real robot works.

SET-UP A target object somewhere across the room, moved between runs.

THE TEST The driver pilots the robot to roughly within arm's reach of the target, then presses a button. From there a program takes over and closes to exactly 10 cm using the ultrasonic. Time the whole thing and measure the final gap.

OUTCOME Faster and more accurate than either a pure driver or a pure program. **This is the literal structure of an FTC match — an autonomous period and a driver period, each doing what it is good at.** It is also how surgical robots and the Mars rovers are operated.